Cardiovascular System (CARD) · NPLEX-P1-CARD-09

CV-9 Blood Distribution to Body Regions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-09-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the aorta and label every branch with its level

A competent response contains

· The ascending aorta gives only the right and left coronary arteries and ends at the sternal angle, the T4 to T5 disc.

· The arch passes back over the left lung root and returns to T4, giving the brachiocephalic trunk, the left common carotid and the left subclavian arteries from right to left.

· The brachiocephalic trunk divides behind the right sternoclavicular joint into the right subclavian and right common carotid arteries.

· Each common carotid divides at about C4, where the carotid sinus holds baroreceptors.

· The descending thoracic aorta runs from T4 to T12 and gives posterior intercostal arteries to spaces 3 to 11, subcostal arteries, and bronchial, esophageal and superior phrenic branches; the first two intercostal spaces are supplied from the costocervical trunk.

· One large anterior segmental medullary artery, usually on the left, reinforces the anterior spinal artery over the lower cord.

· The aorta passes the diaphragm at T12 and bifurcates at L4, near the umbilicus; the median sacral artery leaves the posterior aorta at the bifurcation.

· Abdominal branches with levels: inferior phrenic (paired) about T12 to the diaphragm, celiac trunk T12 to the foregut as far as the major papilla with the liver and spleen, superior mesenteric L1 to the midgut from the distal duodenum to the proximal two thirds of the transverse colon, renal and middle suprarenal (paired) about L1, gonadal (paired) about L2, inferior mesenteric L3 to the hindgut from the distal third of the transverse colon to the anal canal above the pectinate line, and four pairs of lumbar arteries from L1 to L4 to the posterior abdominal wall.

· Paired branches are the inferior phrenic, middle suprarenal, renal, gonadal and lumbar arteries; the three gut arteries and the median sacral artery are unpaired.

· The gonadal arteries arise high because the gonads develop high and then descend.

· The pancreatic head and second part of the duodenum sit at the foregut and midgut junction and receive both celiac and superior mesenteric blood, and the spleen is supplied by the celiac trunk because it forms in the dorsal mesogastrium.

Distinctions that matter

· The arch gives three branches but five arteries reach the head and arms, because the brachiocephalic trunk divides again.

· Levels are not symmetrical about the branch they name: the celiac trunk and the diaphragmatic hiatus are both at T12, while the renal arteries sit at about L1 with the superior mesenteric artery just above them.

Common misconceptions

· Writing that the right subclavian and right common carotid arteries arise from the arch. They arise from the brachiocephalic trunk behind the right sternoclavicular joint.

· Giving the aorta a lumbar origin for the gonadal arteries at the level of the gonads. They arise at about L2, well above the pelvis, because the gonads descend from a high position.

Where it is taught: Section C01, paragraphs on the ascending aorta and arch, the descending thoracic aorta, and the table of abdominal aortic branches of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-BD2

Brain dump 2 · integration and application: Three patients with a lesion on the aorta

A competent response contains

· Patient 1: the left recurrent laryngeal nerve hooks under the arch beside the ligamentum arteriosum, so an aneurysm or mass there stretches it and causes hoarseness.

· Patient 2: the clamp interrupted the large anterior segmental medullary artery, usually on the left, which reinforces the anterior spinal artery over the lower cord.

· Patient 2: the anterior two thirds of the cord infarcts, giving paraplegia with loss of pain and temperature sensation below the lesion.

· Patient 2: vibration and position sense travel in the posterior columns, which are supplied by the posterior spinal arteries and are therefore spared.

· Patient 3: adult coarctation sits just distal to the origin of the left subclavian artery, so both arms are supplied from above the narrowing at high pressure while the lower body is supplied past it.

· Patient 3: the legs are perfused through collateral channels rather than directly, which is why the femoral pulses are weak and delayed compared with the radial pulse.

· Patient 3: notching comes from enlarged, tortuous posterior intercostal arteries eroding the inferior rib borders at about ribs 3 to 8; the arteries of the first two spaces arise from the costocervical trunk above the narrowing, so they are not enlarged and ribs 1 and 2 are spared.

· Levels and landmarks: the arch and the ligamentum arteriosum at T4, the descending thoracic aorta from T4 to T12 for the medullary artery, and the coarctation just distal to the left subclavian origin at the level of the arch.

Distinctions that matter

· Anterior cord infarction and posterior column loss separate on arterial supply, not on level: the anterior spinal artery carries the pain and temperature pathways and the corticospinal tracts, and the posterior spinal arteries carry the posterior columns.

· High arm pressure with weak femoral pulses locates a narrowing between the arch branches and the abdominal aorta; weak femoral pulses with normal arm pressure points instead to iliac or aortoiliac disease.

Common misconceptions

· Blaming hoarseness on the right recurrent laryngeal nerve. The left nerve is the one that loops under the arch, so aortic disease affects that side.

· Expecting rib notching on all ribs in coarctation. The spaces supplied from above the narrowing are not part of the collateral route.

Where it is taught: Section C01, arch paragraph, descending thoracic aorta paragraph, the paragraph on findings predicted by level, and the coarctation flow diagram of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-RT01

Retrieval exercise: Rebuild the collateral route around a coarctation

A competent response contains

· Step 1: pressure in the subclavian artery above the narrowing is high.

· Step 2: the internal thoracic artery descends behind the costal cartilages.

· Step 3: anterior intercostal arteries carry blood around each space.

· Step 4: flow reverses in the enlarged, tortuous posterior intercostal arteries.

· Step 5: blood enters the descending aorta below the coarctation.

· Step 6: the enlarged intercostal arteries notch the inferior rib borders at about ribs 3 to 8.

· The reversed vessel is the posterior intercostal artery; normally it carries blood from the aorta outward, and here the pressure gradient runs from the subclavian system toward the low pressure aorta below the narrowing, so flow runs back toward the aorta.

· Ribs 1 and 2 are spared because their intercostal arteries come from the costocervical trunk, which lies above the narrowing and so carries no collateral load.

Distinctions that matter

· The collateral route is driven by the pressure difference across the coarctation, not by any new vessel: every segment of the path exists normally and simply enlarges.

· Notching is produced by the posterior intercostal arteries, while the internal thoracic and anterior intercostal arteries are the upstream part of the same route.

Common misconceptions

· Drawing flow passing through the narrowing itself. The point of the route is that blood reaches the aorta below the coarctation without crossing it.

· Marking reversed flow in the internal thoracic artery. Flow there runs in its normal direction, downward; the reversal is in the posterior intercostals.

Where it is taught: Section C01, flow diagram on the collateral route around an adult coarctation of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the circle of Willis and its feeding vessels

A competent response contains

· The two internal carotid arteries form the anterior circulation; the two vertebral arteries, each the first branch of the subclavian artery, rise through the transverse foramina from C6 to C1 and join at the lower pons as the basilar artery to form the posterior circulation.

· The internal carotid artery has no branches in the neck, crosses the cavernous sinus beside the abducens nerve, gives the ophthalmic artery, then gives the posterior communicating artery and divides into the anterior and middle cerebral arteries.

· The circle is formed by the anterior communicating artery, the proximal anterior cerebral arteries, the terminal internal carotid arteries, the posterior communicating arteries and the proximal posterior cerebral arteries.

· The middle cerebral artery continues the line of the internal carotid artery and therefore receives most emboli.

· The lenticulostriate branches leave the high pressure middle cerebral trunk at right angles as thin end arteries and supply deep structures including the internal capsule.

· The basilar artery gives the anterior inferior cerebellar and superior cerebellar arteries and then the posterior cerebral arteries; the posterior inferior cerebellar artery arises from the vertebral artery.

· The oculomotor nerve emerges between the posterior cerebral artery and the superior cerebellar artery.

· Territories: anterior cerebral artery to the medial hemisphere with the leg represented more than the arm, middle cerebral artery to the lateral convexity with the face and arm more than the leg and the language areas, posterior cerebral artery to the occipital cortex, posterior inferior cerebellar artery to the lateral medulla, anterior inferior cerebellar artery to the lateral pons.

· The ophthalmic artery supplies the eye, so a carotid embolus can cause transient monocular blindness.

· The external carotid artery supplies the face, scalp and dura but no brain tissue; its one branch that matters inside the skull is the middle meningeal artery, from the maxillary artery, which runs under the pterion and is torn in epidural hematoma.

· Berry aneurysms form at branch points of the circle, about 90% anteriorly and most often at the anterior communicating junction; one at the origin of the posterior communicating artery compresses the oculomotor nerve and dilates the pupil early.

Distinctions that matter

· The circle is an inflow ring: the communicating arteries connect the two carotid systems with each other and with the vertebrobasilar system, while the cortical branches beyond it connect only through small leptomeningeal channels.

· The vertebral arteries give the posterior inferior cerebellar arteries and the basilar artery gives the anterior inferior and superior cerebellar arteries, so keep the parent vessel with each cerebellar artery.

Common misconceptions

· Drawing branches of the internal carotid artery in the neck. It gives none until it is inside the skull.

· Treating the external carotid artery as irrelevant once it supplies no brain. Its middle meningeal branch under the pterion causes epidural hematoma.

Where it is taught: Section C02, paragraphs on the internal carotid and vertebrobasilar systems and on the circle of Willis of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-BD2

Brain dump 2 · integration and application: Place four deficits on the vascular map

A competent response contains

· Patient 1: anterior cerebral artery, anterior circulation; it supplies the medial hemisphere where the leg is represented, along with frontal areas that govern continence and behavior.

· Patient 2: posterior inferior cerebellar artery, posterior circulation; lateral medullary infarction gives vertigo, dysphagia, hoarseness, ipsilateral Horner syndrome, ataxia and ipsilateral loss of facial pain and temperature sensation with contralateral loss over the body.

· Patient 2: findings on opposite sides of the face and the body are crossed findings, which localize the lesion to the brainstem, because the facial sensory nucleus is ipsilateral while the ascending body pathway has already crossed.

· Patient 3: posterior cerebral artery, posterior circulation; homonymous hemianopia with macular sparing follows occipital infarction.

· Patient 4: a lenticulostriate artery, anterior circulation, giving a lacunar infarct in the internal capsule and a pure motor hemiparesis.

· Patient 4: chronic hypertension causes hyaline arteriolosclerosis in these thin end arteries, which also produces lacunar infarcts and Charcot-Bouchard microaneurysms.

· An embolus in a cortical middle cerebral branch still infarcts because the circle of Willis acts only upstream; beyond it, cortical branches connect only through small leptomeningeal anastomoses that cannot carry enough flow.

· The circle protects against slow occlusion of an inflow vessel, and a complete symmetric circle exists in only a minority of people.

Distinctions that matter

· Leg more than arm points to the anterior cerebral artery, and face and arm more than leg to the middle cerebral artery, because of where each territory sits on the motor strip.

· A deep lacunar infarct gives pure motor loss without language or visual field loss, while a cortical middle cerebral infarct adds aphasia or neglect.

Common misconceptions

· Expecting a complete circle of Willis to prevent all stroke. It refills territory only when the block is upstream of the ring and develops slowly.

· Reading vertigo, hoarseness and dysphagia as a cortical problem. The crossed sensory pattern places the lesion in the lateral medulla.

Where it is taught: Section C02, paragraph on the circle of Willis, the table of cerebral and brainstem territories, and the lenticulostriate description of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C03-BD1

Brain dump 1 · foundational retrieval: Limb arteries, their landmarks and their pulse points

A competent response contains

· Upper limb chain: subclavian artery, which passes between the anterior and middle scalene muscles with the brachial plexus, becomes the axillary artery at the lateral border of the first rib, becomes the brachial artery at the lower border of teres major, and divides in the cubital fossa at about the radial neck into the radial and ulnar arteries, which rejoin in the connected superficial and deep palmar arches.

· Lower limb chain: external iliac artery becomes the femoral artery at the inguinal ligament, becomes the popliteal artery at the adductor hiatus, and divides at the lower border of popliteus into the anterior tibial

artery, which pierces the interosseous membrane and continues as the dorsalis pedis artery at the ankle, and the posterior tibial artery, which gives the fibular artery; the plantar vessels rejoin in the plantar arch.

· Pulse points: brachial in the medial arm or in the cubital fossa medial to the biceps tendon, radial lateral to flexor carpi radialis, ulnar near the pisiform, femoral at the midinguinal point just below the inguinal ligament, popliteal deep in the fossa with the knee flexed, dorsalis pedis lateral to extensor hallucis longus, posterior tibial behind the medial malleolus.

· The axillary pulse is not routinely felt; the ulnar and popliteal pulses are often faint; the dorsalis pedis pulse is absent in a small number of healthy adults.

· The connected palmar arches let either the radial or the ulnar artery perfuse the whole hand, which is what the Allen test examines.

· Nerve and artery pairs: the posterior circumflex humeral artery with the axillary nerve around the surgical neck of the humerus, the deep brachial artery with the radial nerve in the radial groove, and the subclavian artery with the brachial plexus between the scalenes.

· Femoral triangle order from lateral to medial: nerve, artery, vein, canal.

· The lateral compartment of the leg has no artery of its own and is supplied by perforating branches of the anterior tibial and fibular arteries.

Distinctions that matter

· A name change is a landmark, not a new vessel: the same column of blood becomes axillary, brachial, then radial and ulnar as it passes fixed structures.

· A faint pulse is not the same as an absent one: the ulnar and popliteal pulses are often hard to feel in healthy people, and only the dorsalis pedis is genuinely absent in some normal adults.

Common misconceptions

· Placing the femoral pulse at the pubic tubercle or in the mid thigh. It lies at the midinguinal point just below the inguinal ligament.

· Treating an absent dorsalis pedis pulse as proof of disease. Check the posterior tibial pulse, because a few healthy adults never have a palpable dorsalis pedis.

Where it is taught: Section C03, paragraphs on the upper and lower limb arteries and the table of limb landmarks and pulse points of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C03-BD2

Brain dump 2 · integration and application: Localize four limb lesions from the examination

A competent response contains

· Patient 1: the first abnormal pulse is the popliteal, so the obstruction lies upstream in the superficial femoral artery, most often in the adductor canal; the calf aches first because claudication appears one level below the narrowing.

· Patient 2: the ankle-brachial index is 70 divided by 140, which is 0.5.

· Patient 2: ankle pressure normally equals or slightly exceeds arm pressure because the pulse wave amplifies as it travels distally, so an index below about 0.9 means pressure has been lost across a stenosis; 0.5 indicates significant disease.

· Patient 3: the brachial artery is injured by a supracondylar humeral fracture, the forearm flexors become ischemic, and the healed result is Volkmann contracture.

· Patient 4: the medial circumflex femoral artery supplies the femoral head, so a displaced intracapsular neck fracture risks avascular necrosis.

· Emboli lodge most often at the common femoral bifurcation in the lower limb.

· Scapular anastomoses bypass an axillary block above the origin of the subscapular artery but not below it, so a proximal block is better tolerated.

· Peripheral arterial disease favors the superficial femoral artery in the adductor canal and the popliteal artery; aortoiliac disease gives buttock and thigh claudication and femoropopliteal disease gives calf claudication.

Distinctions that matter

· Palpating in sequence places an obstruction between the last normal pulse and the first weak one, so the level of the first abnormal pulse is the finding that localizes it.

· Claudication level and pulse level give the same answer from different directions: muscles ache one level below the narrowing while pulses fail at and below it.

Common misconceptions

· Expecting ankle pressure to be lower than arm pressure in health. The pulse wave amplifies distally, so normal ankle pressure matches or exceeds arm pressure.

· Attributing a cold pulseless hand after a supracondylar fracture to the radial artery. The brachial artery is the vessel at that level.

Where it is taught: Section C03, paragraphs on the limb arteries, the peripheral arterial disease paragraph, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw venous return and the portal system

A competent response contains

· The two brachiocephalic veins unite behind the right first costal cartilage to form the superior vena cava, which receives the azygos vein at about T4.

· The inferior vena cava forms at L5, ascends to the right of the aorta, and passes the diaphragm at T8.

· The right gonadal vein drains into the inferior vena cava; the left gonadal vein drains into the left renal vein at a right angle.

· The left renal vein crosses in front of the aorta below the superior mesenteric artery, so it passes between those two vessels; the third part of the duodenum crosses in the same angle.

· The splenic vein and the superior mesenteric vein unite behind the neck of the pancreas to form the portal vein.

· Blood from the gut, spleen, pancreas and gallbladder passes through the liver sinusoids before reaching the inferior vena cava, which is the basis of first pass metabolism.

· The liver receives about a quarter of cardiac output, about three quarters of it portal, yet the portal and arterial sources each supply about half its oxygen.

· Portal pressure is about 9 mm Hg against near 0 mm Hg in the hepatic veins, and the portal system has no valves, so any rise in resistance is transmitted back to the gut and spleen.

· Portosystemic anastomoses: lower esophagus, left gastric vein to esophageal veins draining to the azygos vein, giving varices that can bleed massively; umbilicus, paraumbilical veins to superficial epigastric veins, giving caput medusae; anorectal junction, superior rectal vein to middle and inferior rectal veins, giving anorectal varices; retroperitoneum, colic veins to lumbar and renal veins, giving silent varices.

Distinctions that matter

· The portal system is defined by having a second capillary bed at its far end: gut blood passes the liver sinusoids before any of it reaches the inferior vena cava.

· Portal blood carries most of the liver's flow but only about half its oxygen, so flow share and oxygen share are separate figures.

Common misconceptions

· Drawing both gonadal veins into the inferior vena cava. Only the right does; the left enters the left renal vein.

· Putting the portal vein in front of the pancreas. It forms behind the pancreatic neck from the splenic and superior mesenteric veins.

Where it is taught: Section C04, paragraphs on the superior and inferior venae cavae and the portal system, and the table of portosystemic anastomoses of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-BD2

Brain dump 2 · integration and application: Four patients with an obstructed vein

A competent response contains

· Patient 1: the superior vena cava is obstructed, giving facial, neck and arm swelling with distended chest wall veins.

· Patient 1: if the block lies above the entry of the azygos vein, blood returns through the azygos vein into the superior vena cava below the block; if it lies below that entry, flow reverses in the azygos vein and passes through the lumbar veins into the inferior vena cava.

· Patient 2: the left renal vein is compressed between the superior mesenteric artery and the aorta, the nutcracker phenomenon, causing hematuria and back pressure in the left gonadal vein.

· Patient 2: the left gonadal vein drains into the left renal vein at a right angle and has a longer course, while the right drains directly into the inferior vena cava, which is why varicocele is usually left sided.

· Patient 3: a new isolated right varicocele suggests obstruction of the inferior vena cava or a retroperitoneal mass, because the right gonadal vein drains straight into the cava and has no equivalent compression point.

· Patient 4: the umbilical site connects paraumbilical veins with superficial epigastric veins, giving caput medusae.

· Patient 4: the portal system has no valves, so intrahepatic resistance raises pressure back through the portal and splenic veins and the spleen becomes congested and enlarges; congestive splenomegaly sequesters platelets, which lowers the platelet count.

· Sites of raised portal pressure: prehepatic, for example portal vein thrombosis; intrahepatic, for example cirrhosis, with schistosomiasis acting presinusoidally; posthepatic, for example Budd-Chiari syndrome, right heart failure or constrictive pericarditis.

· The hepatic venous pressure gradient is normally up to about 5 mm Hg, and varices form at about 10 mm Hg.

Distinctions that matter

· Left sided varicocele is explained by venous drainage route and length, while a new right sided varicocele is explained by something obstructing the cava, so the side of the finding changes what you look for.

· Prehepatic, intrahepatic and posthepatic causes all raise portal pressure, but only the intrahepatic and posthepatic groups raise the gradient across the liver.

Common misconceptions

· Expecting a blocked vein to stop venous return. Veins are richly interconnected, so flow reroutes through channels that are normally tiny and then visibly dilate.

· Reading a low platelet count in cirrhosis as bone marrow failure alone. The congested spleen sequesters platelets.

Where it is taught: Section C04, superior vena cava paragraph, inferior vena cava and left renal vein paragraph, portal hypertension classification, and the anastomosis table of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-RT01

Retrieval exercise: Rebuild the cycle that keeps portal pressure high

A competent response contains

· Step 1: fibrosis, regenerative nodules and stellate cell contraction raise intrahepatic resistance.

· Step 2: nitric oxide and other vasodilators that escape the liver dilate the splanchnic arterioles.

· Step 3: portal inflow rises against the high resistance.

· Step 4: low effective arterial volume activates the renin angiotensin aldosterone system and the sympathetic nerves.

· Step 5: the kidneys retain sodium and water, expanding volume.

· Step 6: the extra volume increases splanchnic inflow, which keeps portal pressure high and closes the loop.

· The three intrahepatic changes raising resistance are fibrosis, nodules and stellate cell contraction.

· The mediator dilating the splanchnic arterioles is nitric oxide.

· At step 4 the body senses a low effective arterial volume, because splanchnic dilatation has enlarged the vascular space even though total volume is normal or high.

· Pressure equals resistance times inflow, and cirrhosis raises both.

· The sodium and water retention at step 5 is also what feeds ascites.

Distinctions that matter

· Resistance starts the problem and inflow sustains it, which is why pressure does not settle once the liver is scarred.

· Effective arterial volume and total blood volume are different quantities, and in cirrhosis they move in opposite directions.

Common misconceptions

· Treating portal hypertension as resistance alone. Splanchnic dilatation raises inflow, and pressure is the product of the two.

· Explaining the sodium retention as dehydration. It is a response to a low effective arterial volume in a dilated circulation.

Where it is taught: Section C04, cycle diagram on why portal pressure stays high in cirrhosis of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-CA01

Clinical application: Vomiting blood in a man with long-standing liver disease

A competent response contains

· Recognize: esophageal varices at the lower esophagus, where the left gastric vein meets esophageal veins, and those esophageal veins drain into the azygos vein.

· Explain: the raised resistance lies inside the liver, so portal blood takes the low resistance portosystemic routes upstream of the obstruction instead of crossing the sinusoids to the inferior vena cava, and the valveless portal system transmits the pressure back to the gut and spleen; the distended abdomen with shifting dullness is ascites from sodium and water retention driven by splanchnic dilatation and a low effective arterial volume, and the caput medusae is the umbilical anastomosis between paraumbilical and superficial epigastric veins opening under the same pressure.

· Differentiate: the anorectal anastomosis links the superior rectal vein with the middle and inferior rectal veins and produces anorectal varices, which are not the same as internal hemorrhoids, and hemorrhoids are not more frequent in portal hypertension; bleeding from raised portal pressure arises at a named anastomotic site, comes with splenomegaly, thrombocytopenia and ascites, and follows a raised hepatic venous pressure gradient, whereas a bleed from a gastric wall lesion arises at a single mucosal site with no portosystemic features.

· Investigate: the hepatic venous pressure gradient, which is normally up to about 5 mm Hg and is raised here at 14 mm Hg with varices expected above about 10 mm Hg; portal pressure, normally about 9 mm Hg and raised in this man; hepatic vein pressure, normally near 0 mm Hg; and a platelet count and spleen size, both of which reflect portal congestion.

· Decide: the gradient of 14 mm Hg sits well above the normal ceiling of about 5 mm Hg and above the varix threshold of about 10 mm Hg, and the raised resistance in this man is intrahepatic, from cirrhosis.

· Justify: an intrahepatic block raises portal pressure while hepatic vein pressure stays near 0 mm Hg, so the gradient across the liver is wide; a prehepatic cause such as portal vein thrombosis raises portal pressure with a normal gradient because the liver itself is not the resistance, and a posthepatic cause such as Budd-Chiari syndrome, right heart failure or constrictive pericarditis raises the pressure downstream of the sinusoids as well, so the measured hepatic vein pressure rises with it; the valveless portal system then carries the pressure back to the gut and spleen and opens the anastomoses.

Distinctions that matter

· A raised portal pressure is not the same finding as a raised gradient across the liver, and the difference between them is what separates prehepatic from intrahepatic and posthepatic causes.

· Varices, ascites, splenomegaly and thrombocytopenia are four expressions of the same raised pressure rather than four separate problems.

Common misconceptions

· Calling the bleeding site a hemorrhoidal one. The anorectal anastomosis gives anorectal varices, and hemorrhoids are a separate entity.

· Explaining the ascites as low albumin alone. The notes trace it to sodium and water retention driven by splanchnic dilatation and a low effective arterial volume.

Where it is taught: Section C04, portal system paragraph, portal hypertension classification, the cirrhosis cycle diagram, the anastomosis table, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-BD1

Brain dump 1 · foundational retrieval: Map the watershed zones and explain why they fail first

A competent response contains

· A watershed is tissue at the far edge of two adjacent arterial territories, supplied by the most distal branches of each.

· Pressure falls along every arterial tree because each segment has resistance, so the border receives the lowest perfusion pressure of either tree, and the collaterals there, the leptomeningeal anastomoses in the brain and the marginal artery in the colon, are too small to rescue either side.

· Cerebral blood flow stays nearly constant across a mean arterial pressure of about 60 to 150 mm Hg; below that range flow falls in proportion to pressure and the border zones fail first.

· Chronic hypertension shifts the autoregulatory range upward, so infarction occurs at pressures that would be safe for others.

· Cortical border zones: between the anterior and middle cerebral arteries as a strip over the upper convexity, and between the middle and posterior cerebral arteries in the parieto-occipital region; the internal border zone lies in deep white matter.

· Bilateral anterior to middle cerebral border infarction weakens both proximal arms with the face, hands and legs spared, described as man in a barrel; bilateral middle to posterior cerebral border infarction can produce features of Balint syndrome.

· Colonic sites: the splenic flexure, between the superior and inferior mesenteric arteries, and the rectosigmoid junction; the rectum has additional internal iliac supply and is relatively protected.

· Colonic hypoperfusion causes ischemic colitis with left sided pain and bloody diarrhea, and the mucosa fails first because countercurrent exchange in the villi leaves the tips hypoxic.

· Regions that fail first in low flow: cerebral border zones, at the distal ends of two territories, giving bilateral proximal arm weakness; splenic flexure and rectosigmoid, with weak marginal connections, giving ischemic colitis; subendocardium, farthest from the epicardial arteries and perfused mainly in diastole, giving subendocardial infarction; liver zone 3, last on the sinusoid, giving centrilobular necrosis; renal outer medulla, with a low oxygen tension from vasa recta countercurrent exchange and a high transport demand, giving acute tubular necrosis.

· Only the brain and colon sites are true boundaries between two territories; the other three are the distal ends of a single supply path.

Distinctions that matter

· A border zone between two arterial trees is not the same as the far end of one supply path, although both fail early in low flow.

· Hypoperfusion gives mucosal or mural bowel infarction, while occlusion of a vessel gives transmural infarction.

Common misconceptions

· Expecting autoregulation to protect border zones at any pressure. Once arterioles are maximally dilated, flow follows pressure.

· Assuming a hypertensive patient tolerates low pressure as well as anyone else. The shifted range means injury begins at a higher pressure.

Where it is taught: Section C05, opening paragraph, the cortical and colonic border zone paragraphs, and the table of regions that fail first in low flow of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-BD2

Brain dump 2 · integration and application: Read a pattern of injury back to a mechanism

A competent response contains

· Bloody stools with left sided pain: the colonic border zones at the splenic flexure, between the superior and inferior mesenteric arteries, and the rectosigmoid junction, where the marginal connections are weak; the injury is ischemic colitis, mucosal or mural rather than transmural.

· Weakness of both shoulders and proximal arms with normal grip: the cortical border zones between the anterior and middle cerebral arteries over the upper convexity on both sides, described as man in a barrel.

· Rising aminotransferases: liver zone 3, last on the sinusoid, giving centrilobular necrosis.

· Rising creatinine with granular casts: the renal outer medulla, which has a low oxygen tension from vasa recta countercurrent exchange and a high transport demand, giving acute tubular necrosis.

· What they share: every organ fell below its autoregulatory range at the same time, so the regions that fail first in low flow were injured together, which is global hypoperfusion.

· An embolus or thrombus fills one artery's territory, usually on one side and in one organ, and occlusion gives transmural bowel infarction rather than mucosal injury.

· Strip like infarcts along border zones, often bilateral and in several organs at once, point to global hypoperfusion.

· A severe carotid stenosis already lowers the pressure delivered to that hemisphere, so a modest systemic fall takes the ipsilateral border zones below their threshold while other patients stay within range.

Distinctions that matter

· Number of territories and number of organs is the discriminator: one territory on one side points to an embolus, several border zones in several organs point to a fall in pressure.

· Depth of bowel injury separates the two mechanisms: hypoperfusion injures mucosa or wall, occlusion kills the full thickness.

Common misconceptions

· Reading four organ findings as four unrelated complications. They are one mechanism expressed in the regions that fail first.

· Expecting arm weakness from a border zone infarct to include the hands. The strip over the convexity spares face, hands and legs.

Where it is taught: Section C05, flow diagram from hypotension to watershed infarct, the border zone table, the paragraph on embolus against global hypoperfusion, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-CP01

Comparison table: Global hypoperfusion against occlusion of one artery

A competent response contains

· Mechanism: global hypoperfusion lowers systemic pressure until arterioles are maximally dilated and flow follows pressure; occlusion of a single artery blocks one supply route while systemic pressure is maintained.

· Position within a territory: global hypoperfusion injures the distal border of two territories; occlusion kills the whole territory of the blocked vessel, border and centre alike.

· Sides: global hypoperfusion is often bilateral and roughly symmetrical; occlusion is usually one sided.

· Organs: global hypoperfusion injures several organs at once, in brain, bowel, liver and kidney; occlusion involves one organ.

· Cerebral lesion shape: global hypoperfusion gives narrow strips along the border zones, and bilateral anterior to middle cerebral strips weaken both proximal arms with the face, hands and legs spared; occlusion gives a wedge filling one cortical territory with face and arm or leg weakness according to the artery.

· Bowel wall: hypoperfusion gives mucosal or mural infarction, because the villus tips are hypoxic first; occlusion gives transmural infarction.

· Causes: global hypoperfusion from shock, cardiac arrest, or hypotension during surgery; occlusion from a cardiac embolus, local thrombosis on a plaque, or vasculitis.

Distinctions that matter

· Pressure is adequate in occlusion and inadequate in global hypoperfusion, which is why one produces a single wedge and the other produces thin bilateral strips.

· The mucosa fails first in hypoperfusion because countercurrent exchange in the villi leaves the tips hypoxic, so depth of injury reports mechanism.

Common misconceptions

· Describing a watershed infarct as a small stroke of the same kind as an embolic one. Its position at the far edge of two territories is what names it.

· Expecting one mechanism to explain findings in four organs when the arteries involved are unconnected. Several organs at once means the pressure fell.

Where it is taught: Section C05, flow diagram, border zone table, colonic paragraph, and the final paragraph on embolus against global hypoperfusion of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-BD1

Brain dump 1 · foundational retrieval: Build the flow and extraction table from memory

A competent response contains

· Gut and liver: about 25% of resting cardiac output, extraction about 15 to 20%, flow serving absorption and processing.

· Kidneys: about 20% of output, extraction about 7 to 10%, flow serving filtration.

· Resting muscle: about 15% of output, extraction near the body average at rest and 75 to 90% in exercise, flow serving metabolism.

· Brain: about 15% of output, extraction about 30 to 35%, flow serving a constant demand.

· Heart: about 4 to 5% of output, extraction about 70 to 75%, flow serving metabolism.

· Skin: about 5 to 6% of output, very low extraction, flow serving heat loss.

· Ranked by share: gut and liver, kidneys, then resting muscle and brain, then skin, then heart. Ranked by extraction: heart, brain, gut and liver, kidneys, skin.

· The rankings differ because flow does not simply follow oxygen need: the kidneys and gut use blood for filtration and processing rather than for oxygen, while the heart takes a small share and strips most of the oxygen from it.

· Systemic organs are perfused in parallel at the same aortic pressure, so each organ's own arteriolar resistance sets its share of cardiac output.

· Arterial blood carries about 20 mL O2/dL. For the whole body, 5 L/min with an arteriovenous difference of 5 mL/dL gives about 250 mL O2/min, extraction about 25%, and mixed venous saturation about 75% with a PO2 of about 40 mm Hg.

· Fick principle: oxygen consumption equals blood flow times the arteriovenous oxygen difference; extraction is that difference divided by arterial content.

· Low extraction means reserve, because the organ can take more oxygen from the blood it already receives; high extraction means extra oxygen can come only from extra flow.

Distinctions that matter

· Share of cardiac output and oxygen extraction are independent figures, and the heart sits at opposite ends of the two rankings.

· Flow per gram is a third separate figure: the kidneys have the highest flow per gram of any large organ while resting muscle has one of the lowest.

Common misconceptions

· Assuming the organ with the highest flow uses the most oxygen. Renal flow serves filtration, and renal venous blood stays highly saturated.

· Treating organs as though they were in series. Parallel perfusion at a common pressure is what lets each organ set its own share.

Where it is taught: Section C06, opening paragraphs, the table of resting flow and O2 extraction, and the paragraph on individual organs of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-BD2

Brain dump 2 · integration and application: Work the numbers, then explain two organs under strain

A competent response contains

· Kidney: arteriovenous difference is 20 minus 18.5, which is 1.5 mL/dL; consumption is 11 times 1.5, which is 16.5 mL O2/min; extraction is 1.5 divided by 20, which is 7.5%.

· Heart: arteriovenous difference is 20 minus 6, which is 14 mL/dL; consumption is 2.25 times 14, which is 31.5 mL O2/min; extraction is 14 divided by 20, which is 70%.

· The kidneys receive about five times the coronary flow and use about half as much oxygen, because renal flow serves filtration of plasma while coronary flow serves the metabolism of working muscle.

· If demand doubles, the kidney can double extraction from 7.5% to 15% without any change in flow.

· The heart would need about 140% extraction, which is impossible because it cannot remove more oxygen than the blood delivers, so coronary flow must roughly double instead.

· Coronary flow rises through metabolic vasodilation, mediated by adenosine and nitric oxide, which is why a fixed stenosis that cannot allow that rise produces ischemia.

· The brain is about 2% of body weight, uses about 20% of resting oxygen, has no oxygen store, and loses consciousness within about 5 to 10 seconds of no flow.

· A rise in arterial PCO2 dilates cerebral arterioles and raises flow; hyperventilation lowers PCO2 and constricts them.

· The kidneys filter about 20% of their plasma, so their venous blood stays highly saturated, and their oxygen use tracks the filtered sodium load.

· The coronary sinus is only 25 to 30% saturated at rest, which is the same statement as a resting extraction of about 70 to 75%.

Distinctions that matter

· Extraction has a ceiling and flow does not, which is why a high extraction organ depends on vasodilation while a low extraction organ has reserve in the blood it already receives.

· Oxygen consumption and blood flow are separate quantities linked only through the arteriovenous difference, which is exactly what the Fick principle states.

Common misconceptions

· Reading high renal flow as high renal oxygen use. Renal venous blood returns nearly fully saturated.

· Expecting the heart to meet extra demand by extracting more. At about 70% extraction there is almost nothing left to take.

Where it is taught: Section C06, worked example using the Fick principle, the extraction table, and the paragraph on individual organs of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-PR01

Prediction exercise: Predict the redistribution of cardiac output in hemorrhage

A competent response contains

· Arterial pressure falls and baroreceptor firing falls with it, because the stretch receptors in the carotid sinus and aortic arch are less distended.

· Sympathetic outflow rises, acting on alpha-1 receptors on arteriolar smooth muscle.

· Flow falls through the skin, the splanchnic bed, the kidneys and resting muscle, because alpha-1 mediated constriction raises resistance in those beds.

· Venoconstriction moves blood out of the splanchnic reservoir, the main blood reservoir of the body, toward the heart, which supports venous return.

· Cerebral and coronary arterioles stay dilated, because local metabolic control rather than sympathetic drive sets their diameter.

· The share of cardiac output reaching the brain and heart rises, because those beds keep their resistance low while the others raise theirs.

· Oxygen extraction rises in the constricted beds, since they must take more oxygen from less flow; the kidneys, skin and splanchnic bed have the reserve to do this because their resting extraction is low.

· If low flow continues, those organs are injured: the renal outer medulla gives acute tubular necrosis, liver zone 3 gives centrilobular necrosis, and the colonic border zones give ischemic colitis.

Distinctions that matter

· Low extraction beds are the ones that can be constricted in shock, while the heart depends on flow, so resting extraction predicts which beds are sacrificed.

· Venoconstriction changes venous return by moving stored blood, while arteriolar constriction changes the distribution of whatever output there is.

Common misconceptions

· Expecting sympathetic drive to constrict every bed. Cerebral and coronary arterioles remain under local control, which is how the share shifts toward them.

· Treating the redistribution as harmless because the brain and heart are protected. The constricted beds tolerate only a brief fall, and prolonged low flow injures them.

Where it is taught: Section C06, flow diagram on redistribution of cardiac output in hemorrhage, the extraction table, and the border zone table in C05 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-Q01

Board-style question: Weight loss and vomiting with an aortic landmark

Correct answer: C. Superior mesenteric artery at the level of the first lumbar vertebra

Why it is right: The superior mesenteric artery leaves the aorta at L1 and crosses in front of the third part of the duodenum, so loss of retroperitoneal fat narrows the aortomesenteric angle and compresses the duodenum, with relief when position changes take weight off that angle.

A is wrong: The celiac trunk arises at T12 and supplies the foregut as far as the major papilla; it runs above the duodenum and does not cross its third part.

B is wrong: The inferior mesenteric artery arises at L3 and supplies the hindgut from the distal third of the transverse colon to the anal canal, well below and beyond the duodenum.

D is wrong: The renal arteries arise at about L1 and run laterally to the kidneys rather than across the midline in front of the duodenum.

A competent response contains

· The superior mesenteric artery arises from the aorta at L1 and passes in front of the third part of the duodenum.

· Loss of retroperitoneal fat narrows the angle between the aorta and the superior mesenteric artery, which can obstruct the duodenum, and compression of the left renal vein in the same angle is the nutcracker phenomenon.

Distinctions that matter

· The third part of the duodenum and the left renal vein both cross in the aortomesenteric angle, so one narrowed angle explains two different presentations.

Common misconceptions

· Placing the superior mesenteric artery at T12 with the celiac trunk. The celiac trunk is at T12 and the superior mesenteric artery is at L1.

Where it is taught: Section C01 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-Q02

Board-style question: Dizziness while working with one arm raised

Correct answer: A. Reversal of flow down the left vertebral artery into the dilated arteries of the exercising arm

Why it is right: Arm exercise dilates the arterioles of the arm and lowers pressure further in a subclavian artery already narrowed proximal to the vertebral origin, so flow reverses down the left vertebral artery into the arm and the posterior circulation is drained, which is why vertigo and imbalance appear only during arm use.

B is wrong: Carotid embolization to the middle cerebral artery would produce a sudden fixed deficit of face and arm weakness or language loss on one side, not symptoms that come only while the arm is working.

C is wrong: A cervical rib compressing the subclavian artery between the scalenes affects the arm and the brachial plexus and does not drain the posterior circulation, so it would not cause vertigo.

D is wrong: Occlusion of the posterior inferior cerebellar artery gives a fixed lateral medullary syndrome with dysphagia, hoarseness and crossed sensory loss, and examination between episodes would not be normal.

A competent response contains

· A systolic difference of 30 mm Hg between the arms with a weak delayed pulse on the low side places a stenosis in that subclavian artery.

· The vertebral artery is the first branch of the subclavian artery, so a stenosis proximal to its origin leaves vertebral pressure below basilar pressure and arm exercise reverses flow down the vertebral artery into the arm.

· The posterior circulation loses blood during arm use, giving vertigo and imbalance alongside arm claudication.

Distinctions that matter

· A stenosis proximal to the vertebral origin steals from the posterior circulation, while one distal to it causes arm claudication alone.

Common misconceptions

· Reading dizziness with arm use as a carotid problem. The symptoms are posterior circulation symptoms and they follow the arm, which is what points to the subclavian artery.

Where it is taught: Section C02 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-Q03

Board-style question: Weak arms after a prolonged fall in blood pressure

Correct answer: D. Perfusion pressure at the distal ends of the anterior and middle cerebral territories falling below the autoregulatory range

Why it is right: Pressure falls along every arterial tree, so the distal ends of the anterior and middle cerebral territories sit at the lowest perfusion pressure; once the systemic pressure drops below the autoregulatory range these border zones infarct as narrow strips over the convexity, sparing the face, hands and legs and leaving proximal arm weakness.

A is wrong: Emboli large enough to block both anterior cerebral arteries would infarct the whole medial surface of both hemispheres and weaken the legs more than the arms.

B is wrong: Sagittal sinus thrombosis causes venous congestion with headache, seizures and hemorrhagic infarction rather than narrow symmetrical arterial border strips after an arrest.

C is wrong: Lenticulostriate arteriolosclerosis produces deep lacunar infarcts in the internal capsule with a pure motor hemiparesis on one side, not bilateral cortical strips.

A competent response contains

· After prolonged cardiac arrest, arterioles are maximally dilated and flow falls in proportion to pressure, so perfusion fails first where pressure is lowest.

· The border between the anterior and middle cerebral territories lies over the upper convexity, and bilateral infarction there weakens both proximal arms with the face, hands and legs spared.

Distinctions that matter

· Strips along a border, present on both sides, point to a fall in pressure; a wedge filling one territory points to occlusion of that artery.

Common misconceptions

· Treating bilateral symmetrical infarcts as two separate emboli. Symmetry at the far edge of two territories is the signature of global hypoperfusion.

Where it is taught: Section C05 of the CV-9 Blood Distribution to Body Regions notes.